

GENERAL AUDIENCE ARTICLE

“Major” Indicators of Success: Graduates' Salaries Five Years Out

By Andrea Reed and Kimberly Williams

From the moment students enter school, they're told that a college degree is the key to a successful, high-paying career. But not all degrees lead to the same financial outcomes. What actually determines a student's future salary?

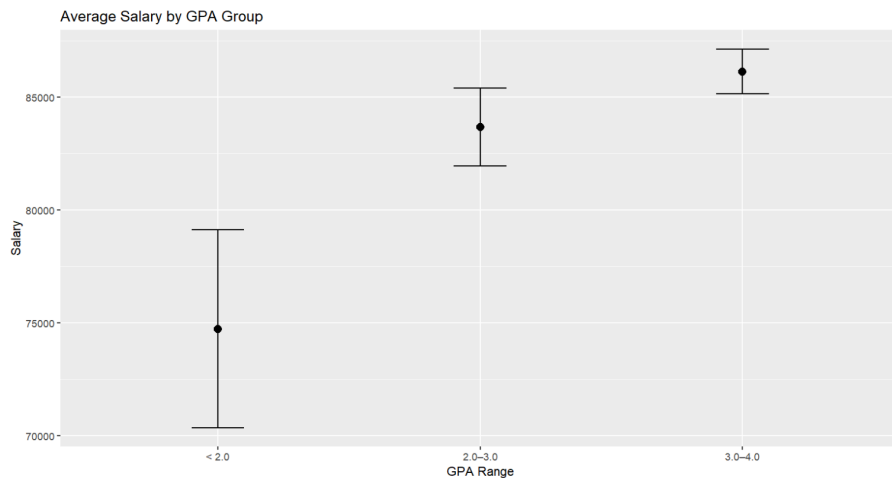
Using data from the American Community Survey Public Use Microdata Series, this article explores how salaries five years post-graduation vary based on a graduate's college major, GPA, and gender. By looking at real earnings data, we aim to shed light on which choices matter most when it comes to pay.



Does Higher GPA lead to Higher Pay?

Do students really need to study? We've all heard that 'Cs get degrees,' but do Cs lead to better salaries down the road? Our studies show that GPA on its own does not impact salary in five years, but GPA does have an effect for different majors.

At a first glance, there is a pattern that suggests higher GPAs, on average, results in students earning more money five years later compared to students with lower GPAs. What is considered 'good enough' when it comes to a student's GPA? When looking at GPAs, those that received lower than 2 have an average salary of \$74734.48, those with a GPA between 2-3 received an average salary of \$83672.83, and those with a GPA of 3 or more received an average salary of \$86132.92. This shows an increasing trend that higher GPA leads to higher pay. However, at a deeper look into the relationship of GPA on salary, we discover that this trend is not strong enough to claim that GPA on its own leads to better salaries.

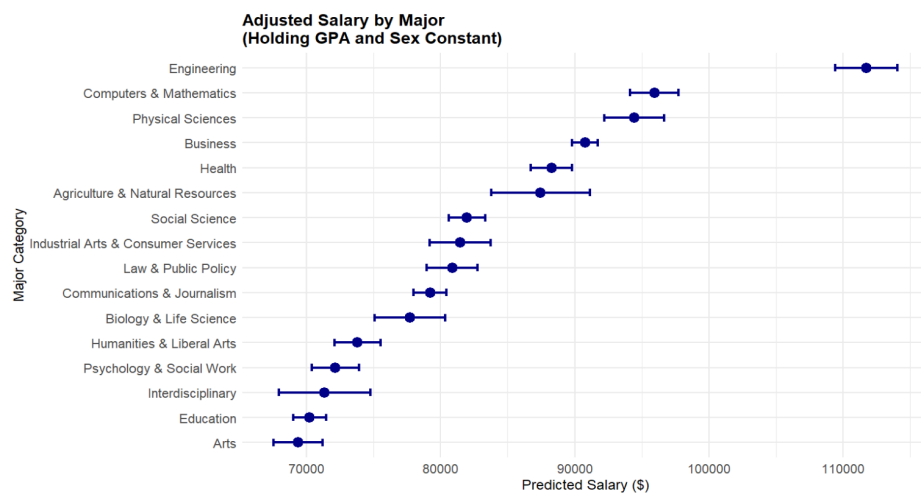


It is important to keep in mind that GPA is not the only factor that determines one's pay. There are some instances where the effect of GPA varies across majors. The effect of GPA is significantly different across majors. Majors, such as business and engineering show that higher GPA is associated with higher salaries. In these fields, a one point increase in GPA results in several thousand dollars of additional annual earnings. Other majors, such as arts and humanities, show a weak relationship between GPA and salary.

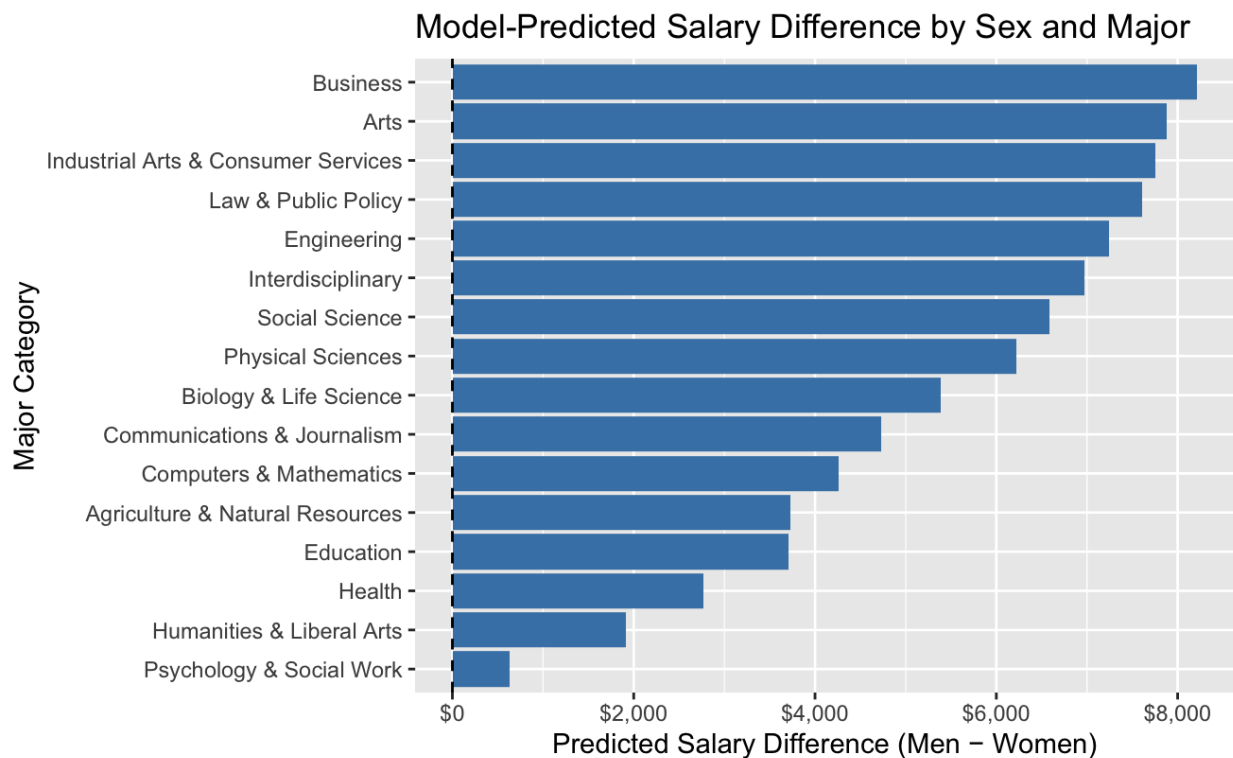
Overall, the results suggest that the general population of students do not need to hyperfocus on their GPA, but higher grades can impact salary within specific fields of study.

Majors that Lead to *Major* Salaries

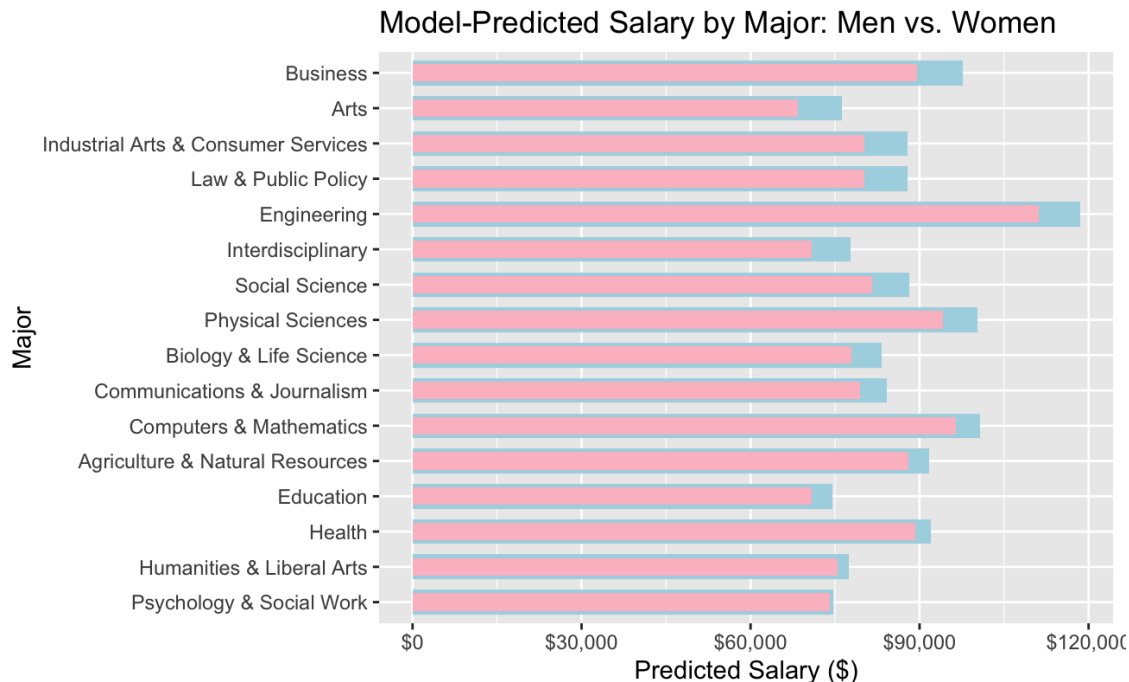
Salaries vary depending on major, with certain majors indicating higher financial success in five years. Holding GPA and Sex constant, Engineering takes the lead for majors that result in larger salaries. A pattern is noticed that indicates majors relating to more technical and mathematical skills result in higher pay than majors relating to social arts. Although there is overlap across major salaries, overall there are significant findings that choice of major will impact future earnings.



Salaries for Men and Women



Salary differences between men and women persist across most college majors. For example, using our model and the average GPA of 3.30207, in majors such as Business and Arts, the gap between male and female earnings exceeds \$7,000 five years after graduation. According to our model, regardless of major, with the average GPA, Men make more than Women. A lower gap doesn't translate to higher pay overall since predicted salaries for women in Engineering still far exceed those in Psychology and Social Work (with the smallest gap). These patterns suggest that while gender differences exist within majors, the choice of major itself remains one of the strongest drivers of early-career salary.



As statistics majors, we were curious what the salary difference would be for us. We found that in the Computers & Mathematics major category, the average salary five years after graduation for men is estimated to be between \$542.89 and \$7329.84 higher than for women.

To get even more specific, as an example, a female in the Computers & Mathematics major category with a 3.8 GPA is predicted to have a 75% chance of making between \$90814.24 and \$103,994.70 annually 5 years post-grad.

Speculation

We've explored how GPA, sex, and major relate to salary, but these factors don't tell the full story. Even after accounting for all three, our model only explained about 78% of the variation in salary 5 years post-grad, so what else could be impacting your pay? Obviously more testing is needed to know for sure, but we can make some guesses:

Professional networks, prior work or internship experience, the university a student attended, and geographic location can all have a substantial impact on pay. Broader economic conditions also matter (graduates entering the workforce during a strong job market may earn more than equally qualified peers graduating during an economic downturn, as seen in 2008).

For example two graduates with the same GPA, sex, and major, can apply for jobs in different areas, one in a high-population metropolis like NYC and another in rural Wyoming, and expect to earn very different wages, highlighting that GPA, sex, and major aren't the only contributing factors to salary.

See the documentation article for a more analytical view on the data, and good luck on the job hunt!

DOCUMENTATION ARTICLE

Modeling:

Our base model is shown below.

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$$

Where $\mathbf{y}$ is the vector of observed salaries and $\mathbf{X}$ is a matrix that includes the explanatory variables (GPA, Major Categories, Sex, interaction between Sex and Major Categories, and interaction between GPA and Major Categories). Beta is the vector of regression coefficients. Major was included as a categorical variable, with Agriculture and Natural Resources serving as the reference group for Major Categories and Female serving as the reference group for Sex.

To assess multiple problems, we fit a linear regression model with salary as the response variable and GPA, MajorCategory, Sex, and the interaction terms as the predictors. Each regression coefficient represents the expected change in salary for a one unit change in the predictor, holding all other variables constant. For the categorical variables, the coefficients represent differences relative to the reference group. The F-test for this model ($F = 62.55$, $p < 2.2e-16$) shows that the model is significantly useful in predicting salary.

We used k-fold Cross Validation to analyze how well our model is at predicting Salaries, and got an RMSE of 5770.659. Since the standard deviation of salary is 12097.75 and RMSE is roughly 47.7% of the standard deviation, we know that our model's typical prediction error is less than half of the natural variability in salaries, indicating that the model captures a substantial portion of the underlying structure in the data.

Process for Impact of GPA:

We interpreted individual coefficients using a t-test and interpreted their p-values. The main effect of GPA across all majors was not statistically significant ($p\text{-value} = 0.69$). It weakly suggested that for a one unit increase in GPA, expected salary is expected to decrease by \$1,934.45.

However, the effect of GPA is found to be statistically significant across individual majors. By looking at the interaction term between GPA * MajorCategory we can see which majors that GPA does play a role in future salary. For example, the interaction effect for Business Majors is 8674.60, meaning that an increase of one point in GPA increases the expected salary by about \$6,740.15 ($= -1934.45 + 8,674.60$).

Process for Salaries Across Majors:

To examine differences in salaries across majors, we predicted salaries from our regression model while holding GPA constant at the mean and sex fixed as Female.

Our model includes both the main effect of major and an interaction between GPA and major and sex and major. This means that the effect of a major on salary depends on a student's GPA and sex. For example, the main effect for Engineering is slightly negative ($-\$2,958.20$), suggesting it might earn less than Agriculture and Natural Resources. However, the interaction

terms are large, so this makes the predicted salaries more large. That's why Engineering appears much higher than other majors in the plot. Students in Engineering are predicted to earn significantly more when they have higher GPAs. Other majors, like Industrial Arts and Consumer Services or Psychology and Social Work, show large negative differences that are statistically significant when looking at the low p-value.

Process for Men and Women:

By looking at the confidence intervals for sex, we were able to assess the role of gender in post graduate salary. Holding GPA and major constant, we are 95% confident that men earn between \$4,768 and \$6,426 more per year than women, on average. This confidence interval shows a significant difference between the pay for men and women because it does not include zero. We also simply looked at average differences by major which show a clear pattern for nearly every major category of men making significantly more than women. In order to get more specific, we also predicted specifically a Computers & Mathematics major category average and individual female with a 3.8 GPA.

Process for Other Factors:

By looking at the Root Mean Squared Error, we can see how well our model predicts data. Our model has an RMSE of 5377.43 which means the predicted salaries are off by about \$5377.43. This shows our model is good, but not the best. Other factors need to be considered as stated above. What could explain salary that we don't have data for? Other explanatory variables, such as location, may be able to improve our RMSE and improve our model fit.

Code:

#Library

```
library(vroom)
library(ggplot2)
library(tidyverse)
library(lmtest)
library(broom)
```

Read in Data and Create Model

```
data <- vroom("Salary.csv")
data_lm <- lm(Salary ~ GPA * MajorCategory + Sex * MajorCategory, data = data)
summary(data_lm)
```

#RMSE

```
preds <- predict(data_lm)
```



```
rmse_model <- sqrt(mean((data$Salary - preds)^2))
rmse_model
```

```
# EDA
```

```
# linearity added variable plot
```

```
car::avPlots(data_lm, terms=~GPA + MajorCategory, ask=FALSE)
```

```
# normality (hist of residuals and KS test)
```

```
standardized_residuals <- MASS::stdres(data_lm)
```

```
ggplot(data = data.frame(x = standardized_residuals),
       mapping = aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 30,
    col = "hotpink",
    fill = "lightpink") +
  stat_function(fun = dnorm,
    args = list(mean = 0, sd = 1),
    color = "darkblue",
    linewidth = 1.2)
```

```
ks.test(x = standardized_residuals,
  y = "pnorm",
  mean = 0,
  sd = 1)
```

```
# equal variance (fitted vs resid and bp test)
library(lmtest)
```

```
# get fitted values and residuals
```

```
fitted_vals <- fitted(data_lm)
```

```
resids <- resid(data_lm)
```

```
# run BP test
```

```
bp_test <- bptest(data_lm)
```

```
# extract p-value and format to 3 decimals
```

```
bp_pval <- round(bp_test$p.value, 3)
```

```
# plot: residuals vs fitted with BP p-value annotation
```

```

ggplot(data.frame(fitted = fitted_vals, resid = resid),
  aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted Values",
    y = "Residuals",
    title = "Residuals vs Fitted Values") +
  annotate("text",
    x = Inf,
    y = Inf,
    label = paste("BP Test p-value:", bp_pval),
    hjust = 1.1,
    vjust = 1.5,
    size = 5)
# independence
# yes each student is independent from one another

```

Cross Validation (K-fold)

```

set.seed(123)
k <- 10
n <- nrow(data)

# Randomly assign each observation to a fold
folds <- sample(rep(1:k, length.out = n))

rmse <- numeric(k)

for (i in 1:k) {
  train_data <- data[folds != i, ]
  test_data <- data[folds == i, ]

  # Fit model on training data
  mod_cv <- lm(Salary ~ GPA * MajorCategory + Sex, data = train_data)

  # Predict on test data
  preds <- predict(mod_cv, newdata = test_data)

  # Compute RMSE
  rmse[i] <- sqrt(mean((test_data$Salary - preds)^2))
}

```

```
# Average RMSE across folds
```

```
mean_rmse <- mean(rmse)
```

```
mean_rmse
```

```
sd_salary <- sd(data$Salary)
```

```
sd_salary
```

```
mean_rmse/sd_salary
```

Look at GPA

```
tidy(data_lm) |> filter(term == "GPA")
```

```
# look at the intercept pvalues
```

```
summary(data_lm)
```

```
# what is a good gpa plot
```

```
data_gpa <- data %>%
```

```
  mutate(GPA_group = cut(GPA,  
                          breaks = c(0, 2, 3, 4),  
                          labels = c("< 2.0", "2.0–3.0", "3.0–4.0"),  
                          include.lowest = TRUE))
```

```
ggplot(data_gpa, aes(x = GPA_group, y = Salary)) +  
  stat_summary(fun = mean, geom = "point", size = 3) +  
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2) +  
  labs(title = "Average Salary by GPA Group",  
       x = "GPA Range",  
       y = "Salary")
```

Look at Majors

```
# compare p vals
```

```
tidy(data_lm) |> filter(grepl("MajorCategory", term))
```

```
# fixed mean gpa and sex(reference group = Female)
```

```
mean_gpa <- mean(data$GPA, na.rm = TRUE)
```

```
data$MajorCategory <- factor(data$MajorCategory)
```

```
# prediction dataset
```

```
pred_data <- data.frame(  
  MajorCategory = levels(data$MajorCategory),  
  GPA = mean_gpa,
```

```

    Sex = "F"
  )

# predict
pred <- predict(
  data_lm,
  newdata = pred_data,
  interval = "confidence"
)

pred_major <- cbind(pred_data, pred)
pred_major

# plot
ggplot(pred_major,
  aes(x = reorder(MajorCategory, fit), y = fit)) +
  geom_point(size = 4, color = "darkblue") +
  geom_errorbar(
    aes(ymin = lwr, ymax = upr),
    width = 0.3,
    color = "darkblue",
    linewidth = 1
  ) +
  coord_flip() +
  labs(
    title = "Adjusted Salary by Major\n(Holding GPA and Sex Constant)",
    x = "Major Category",
    y = "Predicted Salary ($)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 12)
  )
)

# look at Major
tidy(data_lm) |> filter(grepl("MajorCategory", term))

```

```
# Male and Female
```

```
confint(data_lm)["SexM", ]
```

```
# calculate predicted major differences holding GPA constant at the mean GPA
```

```
gpa_ref <- mean(data$GPA, na.rm = TRUE)
```

```
new_data <- expand.grid(  
  GPA = gpa_ref,  
  MajorCategory = levels(data$MajorCategory),  
  Sex = c("F", "M")  
)
```

```
new_data$pred_salary <- predict(data_lm, newdata = new_data)
```

```
gender_diff <- new_data %>%  
  pivot_wider(  
    names_from = Sex,  
    values_from = pred_salary  
  ) %>%  
  mutate(diff = M - F)  
gender_diff
```

```
# visualize the differences
```

```
ggplot(gender_diff,  
  aes(x = reorder(MajorCategory, diff), y = diff)) +  
  geom_col(fill = "steelblue") +  
  coord_flip() +  
  geom_hline(yintercept = 0, linetype = "dashed") +  
  labs(  
    title = "Model-Predicted Salary Difference by Sex and Major",  
    x = "Major Category",  
    y = "Predicted Salary Difference (Men – Women)"  
  ) +  
  scale_y_continuous(labels = scales::dollar_format())
```

```
# another visualization
```

```
pred_sal <- gender_diff %>%  
  arrange(diff) %>%
```

```

mutate(MajorCategory = factor(MajorCategory, levels = MajorCategory))
ggplot(pred_sal, aes(x = MajorCategory)) +
  geom_col(aes(y = M), fill = "lightblue", width = 0.7) +
  geom_col(aes(y = F), fill = "pink", width = 0.5) +
  coord_flip() +
  labs(
    title = "Model-Predicted Salary by Major: Men vs. Women",
    x = "Major",
    y = "Predicted Salary ($)"
  ) +
  scale_y_continuous(labels = scales::dollar_format())

```

```

# example major in computers and math
example_major <- "Computers & Mathematics"

```

```

example_data <- data.frame(
  GPA = 3.8,
  MajorCategory = example_major,
  Sex = c("F", "M")
)

```

```

example_data$pred_salary <- predict(data_lm, example_data)

```

```

example_data %>%
  pivot_wider(names_from = Sex, values_from = pred_salary) %>%
  mutate(diff = M - F)

```

```

cm_data <- data %>%
  filter(MajorCategory == "Computers & Mathematics")
t.test(
  Salary ~ Sex,
  data = cm_data,
  conf.level = 0.95
)

```

```

new_obs <- data.frame(
  MajorCategory = factor("Computers & Mathematics", levels = levels(data$MajorCategory)),

```

```
Sex = factor("F", levels=levels(data$Sex)),  
GPA = 3.80  
)
```

```
predict(data_lm,  
  newdata = new_obs,  
  interval = "prediction",  
  level = 0.75)
```